

1.3 River landforms

The river processes of erosion, transportation, and deposition create different landforms. These landforms typically occur in different parts of the river, according to which processes are dominant.

Potholes, rapids, waterfalls, and gorges

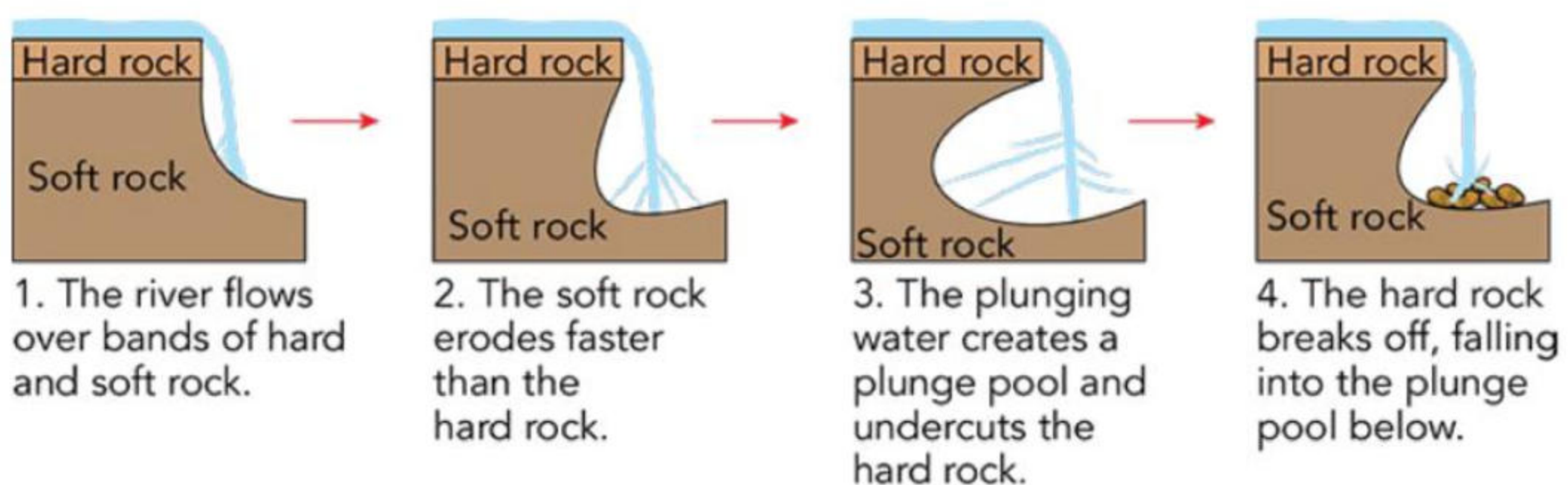
In the upper course of a river, the gradient is steep, and the water is shallow. **Potholes** are cylindrical depressions in the riverbed. They form through the process of abrasion as the material carried by the river scours the riverbed, eroding vertically into the rock. **Rapids** are turbulent sections of a river with fast-flowing water. They form when water encounters obstacles like rocks or alternate bands of hard and soft rock along the riverbed. These obstacles disrupt the flow of the river. This creates turbulence, or whitewater. Higher water levels due to rainfall or snowmelt can make rapids more powerful.



Figure 1.7: River rapids

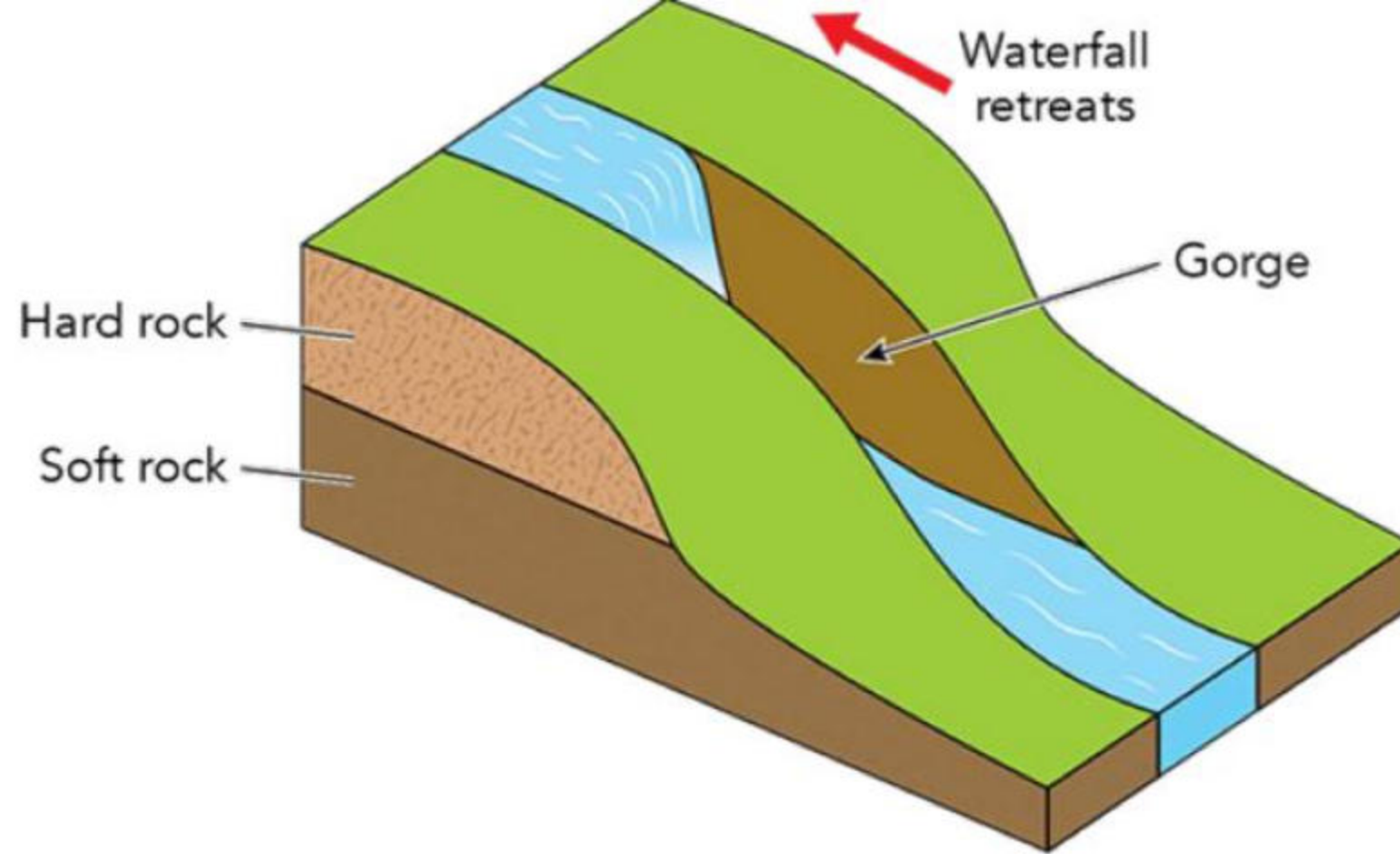
Waterfalls are features where water cascades over a vertical drop in the landscape. They are found where a river flows over bands of hard and soft rock. The river erodes the rock through processes such as hydraulic action and abrasion. The softer rock erodes faster than the harder rock. This creates a drop in the level of the land beneath the river.

When the waterfall is well established, the force of the water plunging over the drop can undercut the hard rock. Eventually the hard rock cannot support itself and it breaks off, falling into the plunge pool below. The force of the falling water, along with the abrasive power of the hard rock fragments, can create a deep plunge pool.



Source A: The formation of waterfalls

This cycle of erosion and undercutting leads to the backward movement upstream, or retreat, of the waterfall. As the waterfall retreats, it leaves a narrow valley. This is called a **gorge**.



Source **B**: A waterfall and gorge

TASK 1: Study Sources **A** **B**

Re-order the statements below to explain how waterfalls and gorges form:

- The cycle of erosion and undercutting leads to the retreat of the waterfall.
- A gorge is created where the waterfall once flowed.
- Water plunges over the edge of the hard rock creating a plunge pool in the soft rock.
- The softer rock becomes lower than the harder rock.
- The hard rock is undercut.
- The hard rock becomes unsupported and breaks off.
- The river flows over bands of hard and soft rock.
- Hydraulic action and abrasion erode the softer band of rock.
- Fragments of hard rock increase rates of erosion in the plunge pool.

Meanders and oxbow lakes

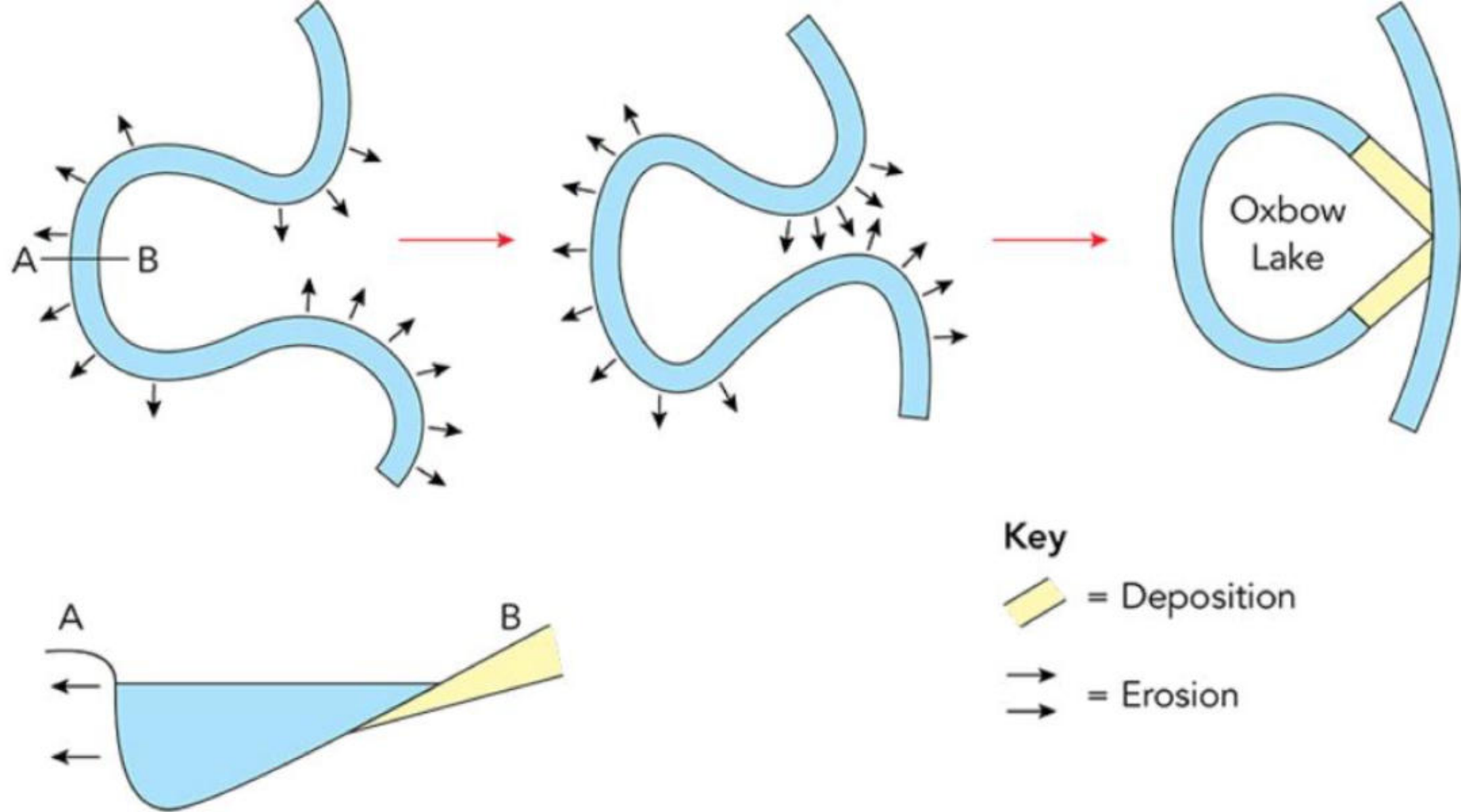
Meanders are curves or bends in a river, typically found in the middle and lower courses of a river where lateral erosion is dominant. The flow of a river is rarely straight because water flows around more resistant parts of the landscape.

As a river winds its way through the landscape, the speed of flow is not even across the channel:

- The outside of the bend has a quicker flow because of **centrifugal forces**, which push water towards the outer bank. This creates energy, increasing rates of hydraulic action and abrasion.
- The inside of the bend has a slower flow and deposition takes place as the river has less energy. Over time, deposited sediment builds up, creating a gently sloping bar.

As the river continues to erode on the outside bank and deposit on the inside bank, meanders become progressively wider.

Oxbow lakes are curved, often u-shaped, bodies of water that are separated from the main river channel. They are the remnants of meander bends that have grown so wide that they become cut off. If the neck of a meander becomes too narrow, the river is able to cut through and follow a shorter route. Water in the old meander bend is called an oxbow lake. Water no longer flows and the lake may fill with sediment and vegetation or dry up completely.



Source **C**: The formation of meanders and oxbow lakes



Source **D**: Meanders and oxbow lakes on the White River in Arkansas, USA

TASK 2: Study Sources **C** **D**

Re-order the statements below. Then, use them to create a flow diagram to explain how meanders and oxbow lakes form:

- This creates deposition on the inside bend.
- The river naturally curves around more resistant features of the landscape.
- The remnant meander becomes an oxbow lake.
- Centrifugal force pushes water to the outside of the curve.
- At the same time, water flows more slowly on the inside of the curve.
- The meander becomes more pronounced over time.
- This increases rates of erosion on the outside bend.

- When the neck of the meander becomes too small, the river breaks through creating a shorter route.

TIP

To remember the location of a river landform, think about the direction of erosion that takes place:

- Vertical erosion, such as with waterfalls, takes place in the upper course.
- Lateral erosion, such as with meanders, takes place in the middle and lower courses.

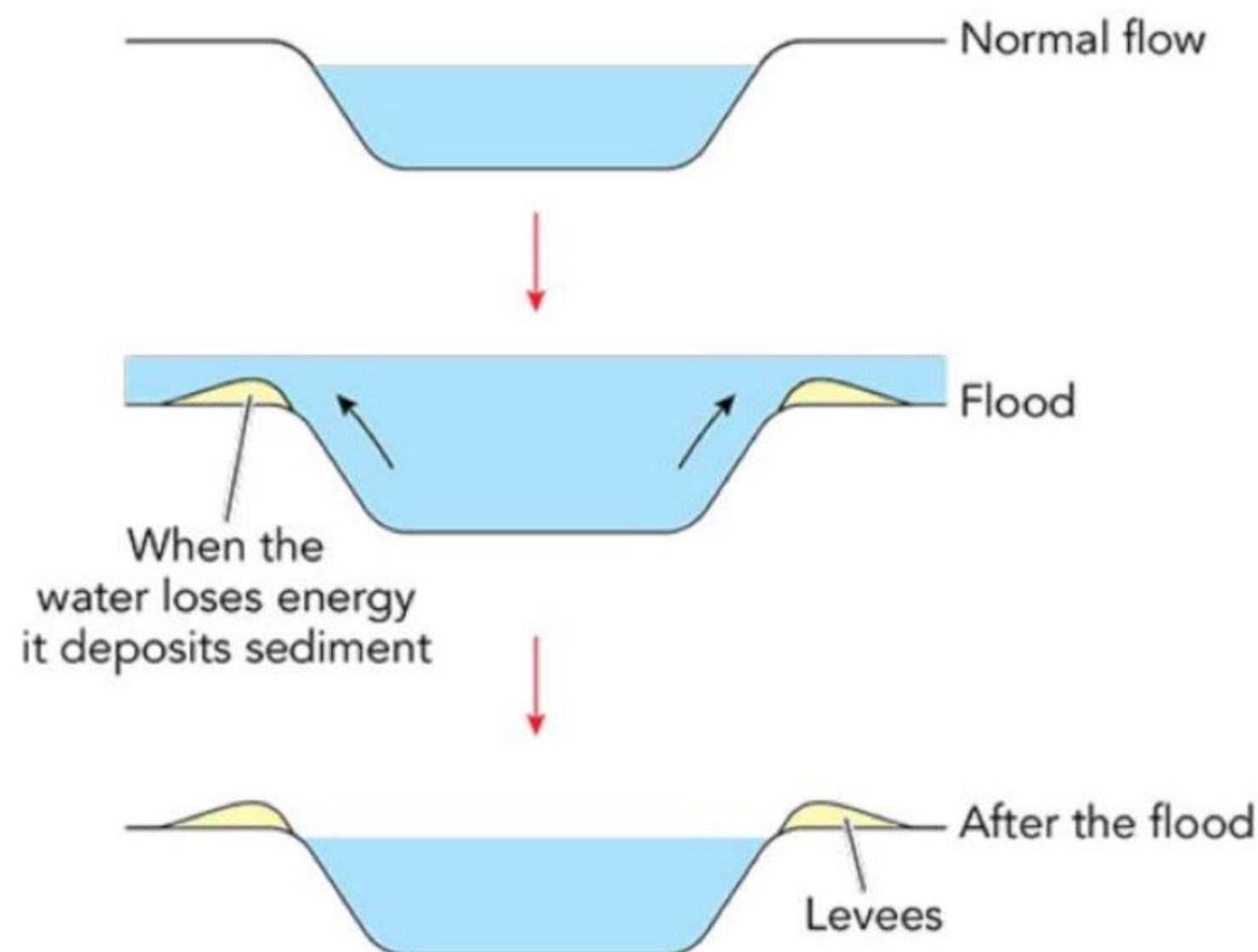
REFLECTION

How can you remember the direction of erosion? Is there anything about each landform that gives a clue to the type of erosion found in that location?

Flood plains and levees

A **floodplain** is an area of flat or gently sloping land beside a river or stream, usually in the middle or lower courses. During times of high discharge, river levels rise and water flows out of the channel and onto the floodplain. As water spreads across the floodplain, it experiences more friction and loses its energy. This leads to the deposition of alluvium. Over time, repeated flooding builds up layers of sediment on the floodplain.

Levees are raised embankments found on floodplains running alongside rivers. As rivers spill onto floodplains, the loss of energy can be almost immediate. This means greater levels of deposition take place at the side of the riverbank. Over time, these layers of deposited sediment build up and become levees, which are higher than the rest of the floodplain.



Source **E**: The formation of levees



Source **F**: A floodplain during times of flood

TASK 3: Study Sources **E** **F**

Work with a partner to create a table of advantages and disadvantages of floodplains and levees.

Deltas

Deltas are landforms at the mouths of rivers, formed through the build-up of deposited material. They often take a shape of a bird's foot (for example, the Mississippi river delta in the USA) or a fan (for example, the Nile river delta in Egypt).

In the lower course, large amounts of sediment are transported by the river. When the river flows into the sea, its velocity decreases dramatically, and it loses the energy it needs to transport its load. Larger, heavier particles require the most energy to transport, and so they are deposited first. Smaller, lighter particles travel further before being deposited. Over time, vast quantities of sediment build upwards and outwards into the sea.

If sediment blocks the main channel, the river divides into **distributaries** (Source G). Bars or islands of land consisting of deposited material are found between the distributary channels.



Source **G**: The Mississippi river delta, USA



Source **H**: The Nile river delta, Egypt

TASK 4: Study Sources **G** **H**

- a** Identify which image shows a fan delta, and which shows a bird's foot delta.
- b** **Make a sketch** of the Mississippi delta and label the following features:
 - a** Main river channel
 - b** Distributaries
 - c** Deposition of finer sediment
 - d** Deposition of larger sediment
 - e** Islands or bars
 - f** The land
 - g** The sea
 - h** The direction of river flow



Source **I**: The River Dove at Fauld, Staffordshire



Source **J**: The River Dove at Milldale, Derbyshire

TASK 5: Study Sources **I** **J**

- Suggest which part of the River Dove is most likely to be in the upper course. Explain why you think this.
- Suggest which part of the River Dove experiences vertical erosion and which experiences lateral erosion. Explain why you think this.
- Identify as many river landforms as you can from Source I.
- Predict how the river in Source I might change over time.